Assignment 3

```
#include <stdio.h>
    void sort(int AT[], int BT[], int Pno[], int size){
        for(int i = 0; i< size; i++){
            for(int j = i+1; j< size; j++){
                if(AT[i] > AT[j]){
                    int temp = AT[i];
                    AT[i] = AT[j];
                    AT[j] = temp;

                    int temp1 = BT[i];
                    BT[i] = BT[j];
                    BT[j] = temp1;

                    int temp2 =Pno[i];
                    Pno[i] = Pno[j];
                    Pno[j] = temp2;
                }
            }
        }
    }

int main() {
    int size;
    printf("Enter the number of processes: ");
    scanf("%d", &size);
    int AT[size], CT[size], TT[size], ST[size], BT[size], WT[size], Pno[size];

    for(int i = 0; i< size; i++){
        Pno[i] = i;
        printf("\nPROCESS %d", i);
        printf("\n");
        printf("\nArrival time: ");
        scanf("%d", &AT[i]);
        printf("\nBurst time: ");
        scanf("%d", &BT[i]);
    }
    for(int i = 0; i< size; i++){

        printf("\nPROCESS %d", i);
        printf("\nArrival time: %d", AT[i]);
        printf("\nBurst time: %d", BT[i]);
    }

    sort(AT, BT, Pno, size);
```

```
    ST[0] = AT[0];
    CT[0] = ST[0] + BT[0];
    for(int i = 1; i< size; i++){
        if(AT[i] > CT[i-1]){
            ST[i] = AT[i];
            CT[i] = ST[i] + BT[i];
        }
        else{
            ST[i] = CT[i-1];
            CT[i] = ST[i] + BT[i];
        }

    }
    printf("\nPno\t\t\tAT\t\t\tBT\t\t\tCT\t\t\tTT\t\t\tWT\n");
    for(int i= 0; i< size; i++){
        TT[i] = CT[i] - AT[i];
        WT[i] = TT[i] - BT[i];
        printf("%d\t\t\t%d\t\t\t%d\t\t\t%d\t\t\t%d\t\t\t%d\n", Pno[i], AT[i], BT[i], CT[i],
TT[i], WT[i]);
    }
    return 0;
}
```

```
input
Enter the number of processes: 4

PROCESS 0

Arrival time: 0

Burst time: 7

PROCESS 1

Arrival time: 2

Burst time: 4

PROCESS 2

Arrival time: 4

Burst time: 1

PROCESS 3

Arrival time: 5

Burst time: 4

PROCESS 0
Arrival time: 0
Burst time: 7
PROCESS 1
Arrival time: 2
Burst time: 4
PROCESS 2
Arrival time: 4
Burst time: 1
PROCESS 3
Arrival time: 5
Burst time: 4
Pno                     AT                      BT                      CT                      TT                      WT
0                       0                       7                       7                       7                       0
1                       2                       4                       11                      9                       5
2                       4                       1                       12                      8                       7
3                       5                       4                       16                      11                      7


...Program finished with exit code 0
Press ENTER to exit console.
```